

RUN EROSION AND EXACT FIBRES FOR A BRANDT SANDWICH MAP ON CYCLIC WORDS

ANONYMOUS

ABSTRACT. Let $B_n = \{0\} \cup [n]^2$ be the aperiodic Brandt semigroup of matrix units. On cyclic words of length m , we study the parallel local map $T(x)_i = x_i x_{i+1} x_i$. A nonzero letter survives one step exactly when its successor is its inverse. We lift this local filter to an all-time normal form: a coordinate survives time t precisely along a forward run of t inverse-compatible edges. It follows that every orbit is eventually fixed, with $1 + n$ fixed words for odd m and $1 + n^2$ for even m . Every pointwise tail is one plus a longest cyclic compatible run; for $n \geq 2$ the sharp maximum is m in odd length and $m - 1$ in even length, while $n = 1$ has maximum $\max\{0, m - 1\}$. Independently, we determine every labelled one-step fibre as a cyclic transfer-matrix trace. Nonzero target letters pin adjacent source letters, reducing the trace to an explicit product of powers of one zero-output matrix across target gaps. We diagonalize that matrix, obtain a closed recurrence for the all-zero fibre, characterize the complete image, and verify total fibre mass. Exact enumeration in 26 finite boxes supplies 1,555,420 author-side regression assertions, not proof or novelty evidence. External ownership remains unresolved, so the manuscript is `HOLD_EXTERNAL`.

1. THE MAP AND ITS SUBTRACTION BOUNDARY

For $n \geq 1$, the aperiodic Brandt semigroup is

$$(1) \quad B_n = \{0\} \cup \{(a, b) : a, b \in [n]\}, \quad (a, b)(c, d) = \begin{cases} (a, d), & b = c, \\ 0, & b \neq c, \end{cases}$$

with zero absorbing. Put $(a, b)^* = (b, a)$ and $0^* = 0$. Brandt semigroups and their identities are established algebraic objects [5]. The word “sandwich” also has an established use for a fixed-element variant $x *_a y = x a y$ [2]; our outer letter varies with the site and is repeated on both sides.

Fix a cyclic length $m \geq 1$, with indices in $\mathbb{Z}/m\mathbb{Z}$. The literal deterministic map on B_n^m is

$$(2) \quad (Tx)_i = x_i x_{i+1} x_i.$$

This is a right-sided nearest-neighbour cellular map on a finite cyclic carrier. Finite cyclic cellular-automaton transformation semigroups are standard background [1]. Explicit iterates for local rules furnished by small associative semigroups are also known [3]. Here the binary local operation $(u, v) \mapsto uvu$ is not asserted to be a semigroup multiplication.

De Bruijn and path automata are standard tools for cellular images and preimages [4]. We assign that machinery, Brandt identities, sandwich terminology, and generic support erosion zero contribution credit. The retained result is the conjunction of a parity-sensitive temporal classification and an explicit every-target inverse factorization for (2). Our owner search is bounded. A search non-hit is not a novelty or priority certificate, and a later literal or equivalent owner requires withdrawal or repositioning.

2. ALL-TIME DYNAMICS

For $x \in B_n^m$, define its cyclic compatibility indicators by

$$(3) \quad g_i(x) = \mathbf{1}\{x_i \neq 0 \text{ and } x_{i+1} = x_i^*\}.$$

Lemma 2.1 (local filter). *For $u, v \in B_n$,*

$$(4) \quad uvu = \begin{cases} u, & u \neq 0 \text{ and } v = u^*, \\ 0, & \text{otherwise.} \end{cases}$$

Proof. The assertion holds when $u = 0$. Let $u = (a, b)$ and $v = (c, d)$. The first product is nonzero exactly when $b = c$; the second multiplication is then nonzero exactly when $d = a$. Together these conditions give $v = (b, a) = u^*$, and the surviving product is $(a, b) = u$. $\square$

Theorem 2.2 (good-run normal form). *For every $t \geq 0$, every $x \in B_n^m$, and every site i ,*

$$(5) \quad (T^t x)_i = \begin{cases} x_i, & \prod_{j=0}^{t-1} g_{i+j}(x) = 1, \\ 0, & \text{otherwise.} \end{cases}$$

The product is empty when $t = 0$.

Proof. At $t = 0$, equation (5) is the identity. Suppose it holds at time t . Lemma 2.1 says that site i survives the next update only if the time- t letters at i and $i + 1$ both survive and are inverses. The induction hypothesis requires the edge blocks $i, \dots, i + t - 1$ and $i + 1, \dots, i + t$ to be good. Their union is $i, \dots, i + t$, and its first edge supplies the inverse relation. When this condition holds, the output remains the original letter x_i ; otherwise it is zero. This proves the claim at $t + 1$. $\square$

Corollary 2.3 (fixed states). *Every recurrent state is fixed, and*

$$(6) \quad |\text{Fix}(T)| = \begin{cases} 1 + n, & m \text{ odd,} \\ 1 + n^2, & m \text{ even.} \end{cases}$$

Proof. Theorem 2.2 makes the nonzero support nonincreasing and sends every non-all-good word to zero after at most m steps, so a recurrent word is fixed. A fixed word with one nonzero letter must have every successor equal to the inverse of its predecessor; hence all letters are nonzero and alternate under $*$. Odd length forces the starting unit to be diagonal, giving n choices, whereas even length permits all n^2 units. The all-zero word supplies the additional fixed point. $\square$

For a non-all-good cyclic binary word, let $L(x)$ denote the length of its longest cyclic run of good edges. Define the tail $\mu(x)$ as the least $t \geq 0$ for which $T^t x$ is fixed.

Theorem 2.4 (pointwise and sharp tails). *The all-zero and all-good words have tail zero. Every other word satisfies*

$$(7) \quad \mu(x) = L(x) + 1.$$

For $n \geq 2$,

$$(8) \quad \max_x \mu(x) = \begin{cases} m, & m \text{ odd,} \\ m - 1, & m \text{ even.} \end{cases}$$

For $n = 1$, the maximum is $\max\{0, m - 1\}$.

Proof. By Theorem 2.2, some nonzero coordinate remains at time $L(x)$, while no coordinate remains at time $L(x) + 1$. This proves (7).

Assume $n \geq 2$ and choose an off-diagonal unit u . When m is odd, alternate u, u^* along $m - 1$ successive edges. The closing edge is the unique bad edge, so $L = m - 1$. When m is even, a word cannot have exactly one bad edge: traversing the other $m - 1$ inverse relations forces the omitted relation as well. Thus $L \leq m - 2$. Alternating along $m - 2$ edges and then repeating u produces two adjacent bad edges and attains this bound.

If $n = 1$, write e for the unique nonzero unit. The good edges are exactly the occurrences of ee . A mixed word has $L \leq m - 2$, with equality for one zero followed cyclically by $m - 1$ copies of e when $m \geq 2$. For $m = 1$, both words are fixed. $\square$

Remark 2.5 (the short cycles). For $m = 1$, the fixed states are 0 and the n diagonal units. The maximum tail is one for $n \geq 2$ and zero for $n = 1$. For $m = 2$, the nonzero fixed words are (u, u^*) , one for each nonzero unit; every other word maps to $(0, 0)$ in one step. These cases agree with Corollary 2.3 and Theorem 2.4.

3. EVERY-TARGET FIBRES AND THE IMAGE

Put $Q = B_n$, $q = n^2 + 1$, and for each output letter $y \in Q$ define the $q \times q$ zero-one matrix

$$(9) \quad M_y(u, v) = \mathbf{1}\{uvu = y\} \quad (u, v \in Q).$$

Rows represent the current source letter and columns the next source letter.

Theorem 3.1 (labelled trace formula). *For every target $y = (y_0, \dots, y_{m-1}) \in Q^m$,*

$$(10) \quad |T^{-1}(y)| = \text{tr}(M_{y_0} M_{y_1} \cdots M_{y_{m-1}}).$$

Proof. Expanding the trace gives a sum over cyclic source-letter paths $u_0, \dots, u_{m-1}$ of $\prod_i M_{y_i}(u_i, u_{i+1})$. A product equals one exactly when $u_i u_{i+1} u_i = y_i$ at every site. These paths are precisely the source words in the labelled fibre. $\square$

Let $A = M_0$. Suppose a target has nonzero sites $i_1, \dots, i_s$ in cyclic order. Let h_j be the number of zero target sites strictly between i_j and i_{j+1} , with $i_{s+1} = i_1$ cyclically.

Theorem 3.2 (nonzero-anchor gap product). *If $s \geq 1$, then*

$$(11) \quad |T^{-1}(y)| = \prod_{j=1}^s (A^{h_j})_{y_{i_j}^*, y_{i_{j+1}}}.$$

For $y = 0^m$, the fibre is $\text{tr}(A^m)$.

Proof. By Lemma 2.1, a nonzero output y_i pins the adjacent source pair to (y_i, y_i^*) . Between two pinned pairs, every zero output inserts one transition of A . Multiplying the path counts across all cyclic gaps gives (11). With no pinned site, the cyclic path expansion is exactly $\text{tr}(A^m)$. $\square$

The zero-output trace admits a closed form. Put $r = n^2$, and define

$$(12) \quad s_0 = 2, \quad s_1 = r, \quad s_k = r s_{k-1} + s_{k-2} \quad (k \geq 2).$$

Theorem 3.3 (zero-fibre spectrum). *For every $m \geq 1$,*

$$(13) \quad |T^{-1}(0^m)| = s_m + (-1)^m \left(\frac{r+n}{2} - 1 \right) + \frac{r-n}{2}.$$

The spectrum of A consists of the two roots of $z^2 - rz - 1$, together with -1 of multiplicity $(r+n)/2 - 1$ and $+1$ of multiplicity $(r-n)/2$.

Proof. The zero row of A is all ones. A nonzero row indexed by u is all ones except for a zero in column u^* . Let P be inversion on the r unit coordinates. On the unit-coordinate subspace whose coefficients sum to zero, $A = -P$. Inversion has n fixed diagonal units and $(r-n)/2$ two-element orbits. Its positive and negative eigenspaces therefore have dimensions $(r+n)/2$ and $(r-n)/2$. Removing the all-unit vector from the positive eigenspace gives the asserted multiplicities for A .

Let e_0 be the zero-coordinate vector and w the sum of all unit-coordinate vectors. Then

$$Ae_0 = e_0 + w, \quad Aw = re_0 + (r-1)w.$$

The matrix on $\text{span}\{e_0, w\}$ has trace r and determinant -1 , hence characteristic polynomial $z^2 - rz - 1$. The sum of the m th powers of its roots obeys (12). Adding the powers of the remaining eigenvalues proves (13). $\square$

Corollary 3.4 (complete image criterion). *The all-zero target lies in $\text{im } T$. A target with nonzero anchors lies in $\text{im } T$ if and only if every consecutive cyclic anchor pair satisfies*

$$(14) \quad \begin{cases} y_{i_{j+1}} = y_{i_j}^*, & h_j = 0, \\ y_{i_{j+1}} \neq y_{i_j}^*, & h_j = 1, \end{cases}$$

with no restriction when $h_j \geq 2$.

Proof. In (11), A^0 is the identity. For $h = 1$, the entry from y^* to z vanishes exactly when $z = y$. For $h \geq 2$, a path can move from any letter to zero, remain there for $h-2$ steps, and then move to any letter, so every entry of A^h is positive. The all-zero source maps to the all-zero target. $\square$

Corollary 3.5 (mass conservation). *The complete labelled fibres satisfy*

$$(15) \quad \sum_{y \in Q^m} |T^{-1}(y)| = q^m.$$

Proof. Every ordered pair (u, v) has one local output, so $\sum_{y \in Q} M_y = J_q$, the all-ones matrix. Summing (10) over all targets gives $\text{tr}(J_q^m) = q^m$. $\square$

The short-cycle fibres are now explicit. At $m = 1$, the zero fibre has $n^2 - n + 1$ sources, each diagonal target has one source, and each off-diagonal target is empty. At $m = 2$, the zero fibre has $(n^2 + 1)^2 - n^2$ sources, while every nonzero alternating target (u, u^*) has its unique source and all other targets are empty.

4. EXACT PRESSURE, LIMITATIONS, AND CONCLUSION

The paper-local standard-library verifier enumerates every source and target in 26 boxes, including $n = 1$ through length ten, $n = 2$ through length seven, and $n = 3, 4, 5$ at shorter lengths. It compares literal iterates with Theorem 2.2, reconstructs every target by a cyclic path dynamic program, attacks matrix direction by dense products, and checks integer eigenspace bases. Selected results appear in Table 1.

n	m	states	image	fixed	zero fibre
1	10	1024	277	2	123
2	5	3125	183	3	1363
2	7	78125	1459	3	24475
3	4	10000	226	10	6895
4	3	4913	89	5	4141

TABLE 1. Selected exhaustive author controls. Enumeration is regression pressure; the proofs cover every $n, m \geq 1$.

The paper treats the aperiodic Brandt semigroup and a cyclic one-sided neighbourhood only. It gives no theorem for Brandt semigroups over nontrivial groups, other boundary conditions, asynchronous schedules, or time- t inverse fibres. The finite controls are not experiments, proofs, or owner evidence. The exact contribution is the closed temporal and one-step inverse package for (2); generic erosion and transfer-matrix vocabulary remain background. External circulation is blocked while ownership remains amber.

DECLARATIONS

Data availability. No external data were used. Exact source code and its canonical stdout accompany the manuscript.

Ethics. The work studies finite algebraic maps and raises no human-subject, personal-data, or deployment issue.

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